INCH-POUND

MIL-A-53089
AMENDMENT 2
30 November 1990
SUPERSEDING
INT. AMENDMENT 1
1 March 1990

MILITARY SPECIFICATION

AIR CONDITIONERS: VERTICAL AND HORIZONTAL, COMPACT, MULTIPLE POWER INPUT

This amendment forms a part of MIL-A-53089, dated 9 February 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2, after line 5 (type I), add:

"D - 36,000 Btu/hr - 208 V, 3-phase, 50/60/400 Hz."

After line 9 (type II), add:

"C - 36,000 Btu/hr - 208 V, 3-Phase, 50/60/400 Hz."

Delete lines 10 through 15 in their entirety.

PAGE 2

2.1.1, line 7, add:

"SPECIFICATIONS

FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
NN-P-71	- Pallets, Material Handing, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall."

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PAGE 3

* 2.1.1, under STANDARDS, MILITARY, delete MIL-STD-781 in its entirety.

2.1.2, after line 13, add:

"TA13229E6700 - Air Conditioner, Horizontal, Compact, 36,000 Btu/hr, 208 Volt-AC, 3-Phase, 50/60/400 Hertz.

* "TA13229E8100 - Air Conditioner, Vertical, Compact, 36,000 Btu/hr, 208 Volt-AC, 3-Phase, 50/60/400 Hertz."

PAGE 4

* 2.2, line 23, ASHRAE STD 16, delete the title in its entirety and substitute "Methods of Testing for Rating Room Air Conditioners and Packaged Terminal Air Conditioners".

PAGE 5

* 2.2, lines 2 and 3, insert "AWS" in front of D1.1 and D1.2.

* 3.1, after line 4 (type I), add

"Size D - 36,000 Btu/hr - TA13229E8100."

After line 6 (type II), add:

"Size C - 36,000 Btu/hr - TA13229E6700"

PAGE 6

* 3.2, delete in its entirety and substitute:

"3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3."

* 3.3.2.1, line 2, delete "when tested as specified in 4.5.2.21".

* 3.3.2.2, lines 2 and 3, delete "when tested as specified in 4.5.2.21".

3.4, line 3, add: "All rubber components shall be ozone resistant and no more than four calender quarters old from the date of manufacture to the date of acceptance by the Government."

PAGE 7

* 3.5.1.2.1, line 1, delete "Type I, size C. Type I, size C units shall" and substitute "Coil frost. The air conditioner shall".

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3.5.1.2.2, delete in its entirety and substitute:

"3.5.1.2.2 Low temperature operation. After being subjected to 70 ±5 °F on the front and 0 ±2 °F for type I, size D and type II, size B, or -25 +0, -2 °F for type II, size C, on the rear for a period of 10 hours, the air conditioners shall start and operate in the cooling mode as specified with air at 70 ±5 °F entering the evaporator and air at 0 ±2 °F for type I, size D and type II, size B, or -25 +0, -2 °F for type II, size C, entering the condenser."

3.5.2, table I, delete in its entirety and substitute:

"TABLE I. Cooling capacities.

Unit	Condition	Ambient temperature (dry bulb)	Evaporator inlet temperature		Minimum net sensible cooling capacity	Minimum net total cooling capacity
			dry bulb	wet bulb		
		°F	°F	°F	Btu/hr	Btu/hr
Type I						
Size C	A	120	90	75	12,100	18,500
	B	95	80	67	14,000	21,500
* Size D	A	120	90	75	22,680	37,800
*	B	95	80	67	24,570	37,800
Type II						
Size B	A	120	90	75	14,000	18,500
	B	95	80	67	14,700	19,500
Size C	A	120	90	75	30,000	41,000
	B	95	80	67	32,000	40,000

Note: Data shown for 0-inch water gage (WG) (free air) external static pressure on the evaporator for type I units and 0.25 inch WG for type II units."

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3.5.3, table II, delete in its entirety and substitute:

"TABLE II. Condensing and suction pressures (psig).

Unit	Operating conditions (see 3.5.1.1)			Operating conditions (see 3.5.2)	
	Condensing pressure (max)	(min)	Suction pressure (min)	Condensing pressure (max)	Suction pressure (min)
Type I					
Size C	470	-	50	440	65
* Size D	470	105	50	430	65
Type II					
Size B	470	160	50	425	50
Size C	470	160	50	430	50

"

3.5.6, table III, delete in its entirety and substitute:

"TABLE III. Airflow, heating capacity and power consumption.

Unit	Evaporator Airflow (minimum SCFM) (1)	Heating (minimum Btu/hr) (2)	Power Consumption (maximum KW) (3)	Power Factor (minimum) (4)
Type I				
Size C	570	12,000	6.5	0.9
* Size D	1,250	28,600	11.0	0.83
Type II				
Size B	580	13,400	6.5	0.9
Size C	1,380	31,200	13.5	0.9

"

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* 3.5.7, after line 4 (type I), add:

"Size D - 208 V, 3-phase, 50/60/400 Hz."

After line 7, (type II), add:

"Size C - 208 V, 3-phase, 50/60/400 Hz.

Total RMS current to ground for type II, size C unit shall not exceed 10mA at full operation when operating at rated voltage."

PAGE 10

* 3.6.1 and table IV, delete in their entirety and substitute:

"3.6.1 Noise limits. Noise levels produced by the air conditioners shall comply with the requirements of MIL-STD-1474, with the exception of 5.2, 5.3, 5.4, and except as indicated herein.

a. The noise level on the evaporator side of the air conditioners shall not exceed MIL-STD-1474, category F [65 dB(A)] at one meter hemispherical perimeter from the center of the evaporator surface. Alternate preferred speech interference levels (PSIL) shall not apply.
b. The noise level on the condenser side of the air conditioners shall not exceed MIL-STD-1474, table IV, category D [85 dB(A)] at a one meter hemispherical perimeter from the center of the condenser surface of the air conditioner. Octave band limits of MIL-STD-1474 shall not apply.
c. Where the noise limits specified above can be documented as being clearly beyond the state-of-the-art, selection of another noise limit shall be made by the procuring activity."

PAGE 12

3.9.4, table V, delete in its entirety and substitute:

"TABLE V. High and low cutout pressures.

Unit	Compressor high-pressure cutout (±10 psig)	Compressor low-pressure cutout (±5 psig)
Type I		
Size C	480	7
* Size D	480	7
Type II		
Size B	480	7
Size C	480	15

"

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3.10.2, line 2, delete: "of the air condition".

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3.15, table VII, delete in its entirety and substitute:

"TABLE VII. Weight.

Type	Size	Maximum Weight (lb)
I	C	280
	D	460
II	B	290
	C	435

"

PAGE 16

* 3.17.1, line 5, delete "inches with" and substitute "inches wg with".

* 3.17.3, delete in its entirety and substitute:

"3.17.3 Block off assembly. When specified (see 6.2), a block-off assembly shall be provided to cover the opening when the control box is removed for remote operation (see 3.10.1)."

PAGE 17

* 3.20.4, line 16, delete "Welding Qualifications" and substitute "Welding and Brazing Qualifications".

PAGE 18

4.1.2, delete in its entirety.

PAGE 19

* 4.3.1. lines 3 and 4, delete "and cause for performing the inspection specified in 4.1.2 as applicable".

* 4.3.2, lines 3 and 4, delete "and performing the inspection specified in 4.1.2" and insert a period at the end of the sentence.

PAGE 21

* Table VIII, column 10 (Requirement paragraph), make the following changes:

Test req 2 (Examination), after "3.18", add "3.19, 3.20".
Test req 3 (Controls), after 3.10.4", add "3.9.2".
Test req 8 (Heating capacity), after "3.5.5.1", add "3.5.6".
Test req 10 (High temperature operation, cooling), after "3.5.1.1", add "3.5.3".

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* 4.5.2.1, delete table title "TABLE VII Test schedule." and substitute "TABLE VIII. Test schedule. (continued)".

After test req 26, add:

"27	- X -	Electrical power supply."	4.5.2.27	3.5.7

* 4.5.2.3.1, line 4, between "Reduce the pressure" and "300 ±5 psig", add "to".

PAGE 27

* 4.5.2.11, line 2, after "temperature of", add "not less than".

* 4.5.2.12.1.1, delete in its entirety and substitute:

"4.5.2.12.1.1 Coil frost. Operate the air conditioner continuously for 12 hours in the cooling mode in an ambient temperature of 55 ±5 °F, with air at +70 ±2 °F dry bulb, and 60 ±3 °F wet bulb, entering the evaporator. Immediately following the 12-hour operation, inspect the evaporator coil. Any evidence of ice or frost on the finned area of the evaporator coil, or failure of the air conditioner to operate and provide cooling for 12 hours shall constitute failure of this test.

* 4.5.2.12.1.2, delete in its entirety and substitute:

* "4.5.2.12.1.2 Low temperature operation. Place the air conditioner in a test chamber, installed for in-the-wall operation, with the air conditioner front exposed to a temperature of 70 ±5 °F and the rear (external portion) exposed to
* 0 ±2 °F for type I, size D and type II, size B, or -25 +0, -2 °F for type II, size C. With power connected to the air conditioner, maintain the temperatures for a 10-hour soak period. Then, set the unit for the coolest temperature, start and operate the air conditioner in the cool mode for a period of not less than 1 hour with the air at 70 ±5 °F entering the evaporator, and the air at 0 ±2 °F for
* type I, size D and type II, size B, or -25 +0, -2 °F for type II, size C, entering the condenser. One or more of the following shall constitute failure of this test:

a. Inability of the air conditioner to start and operate in the cooling mode.
b. Temperature differential between the average entering air temperature and average leaving evaporator air temperature less than 15 °F when operating in the cooling mode after stabilization of conditions has occurred.

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* 4.5.2.12.2, lines 1 and 2, delete "After completion of 4.5.2.12.1,".

* 4.5.2.13, lines 1 and 2, after "temperature of" add "not greater than".

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* 4.5.17.1, renumber as "4.5.2.17.1".

PAGE 35

4.5.2.19.4, table XIII, delete in its entirety and substitute:

"TABLE XIII. Minimum temperature drop and maximum current draw. 1/

Unit	Input Power	Temperature difference °F (minimum)	Current draw Amps (max)
Type I			
C	208V, 3 phase, 60 Hz	20.0	21
* D	208V, 3 phase, 60 Hz	18.2	32
Type II			
B	230V, single phase, 60 Hz	20.0	35
	208V, 3 phase, 60 Hz	20.0	21
C	208V, 3 phase, 60 Hz	20.0	45

1/ For all 3-phase air conditioners, amperage reading shall be the average of the three phases."

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* 4.5.2.23.2.1, line 5, delete "specified for table VI" and substitute "specified in table XIII".

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* 4.5.2.26, delete in its entirety and substitute:

"4.5.2.26 Noise level test. Noise level measurements shall be taken at the evaporator and condenser outlets in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when the air conditioner is operating at full capacity. For air conditioners which use multi-speed fans, noise measurements shall be made with the fan operating at both the highest and lowest speeds. Measurements shall be taken at each outlet, in a horizontal plane which is even with the center of the outlet. In the evaporator section, readings shall be taken directly in line with the center of the outlet and at angles 30 degrees to the left and to the right of the center reading. All noise measurements shall be taken at a distance of one meter from the center of the outlet. Condenser readings shall be performed like evaporator readings with the exception of the angle being 45 degrees. Noise level measurements shall be taken in units of dB(A). Failure to comply with 3.6.1 and MIL-STD-1474 provisions shall constitute failure of this test."

After 4.5.2.26, add:

"4.5.2.27 Electrical power supply. To determine RMS current to ground for the unit, place the air conditioner on a non-conductive surface. Turn power on and measure voltage of each phase with multimeter/voltmeter and record data. Next, attach one of the test lead from digital multimeter (Fluke Model 8050A or equal) to the ground stud of the unit and the other lead to the earth (protective) ground. Set the digital multimeter to indicate RMS current rate of 20mA. Turn the unit "ON" to cool position and operate the unit for 15 minutes. The ambient temperature during operation of the unit shall be between 80 and 95 °F. Monitor the digital multimeter to ensure that RMS current is within specified limits. Record data at 5-minute intervals. Shut the unit off and wait for 5 minutes. Repeat this test one more time. Nonconformance to 3.5.7 shall constitute failure of the test.

* 5.2.1.2, line 1, delete the title and substitute: "Air conditioners."

* 5.3.1.1, line 1, delete the title and substitute: "Air conditioners."

* 5.3.2.1, line 1, delete the title and substitute: "Air conditioners."

* 6.2.d, delete in its entirety and substitute:

"d. When a block-off assembly is required (see 3.17.3)."

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* 6.9, under "Type I - Vertical", add:

"D 1 - 208V, 3-phase, 50/60 Hz D - 208V, 3-phase, 50/60/400 Hz
2 - 208V, 3-phase, 400 Hz"

Under "Type II - Horizontal", add:

"C 1 - 208V, 3-phase, 50/60 Hz C - 208V, 3-phase, 50/60/400 Hz.
2 - 208V, 3-phase, 400 Hz"

PAGE 45

Figure 1, delete lines which read:

"II A 10.25 ±0.5"

and

"II D 15.75 ±0.5"

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